

Exp No:15

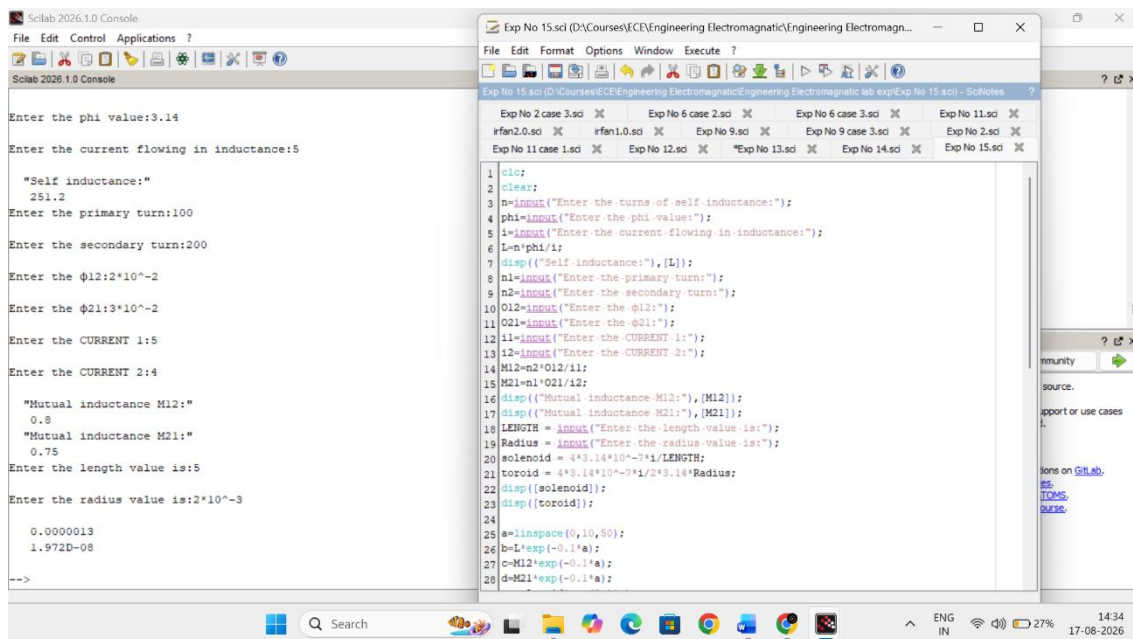
Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Aim:

To determine and plot Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Software required:

SCILAB version 6.0.1



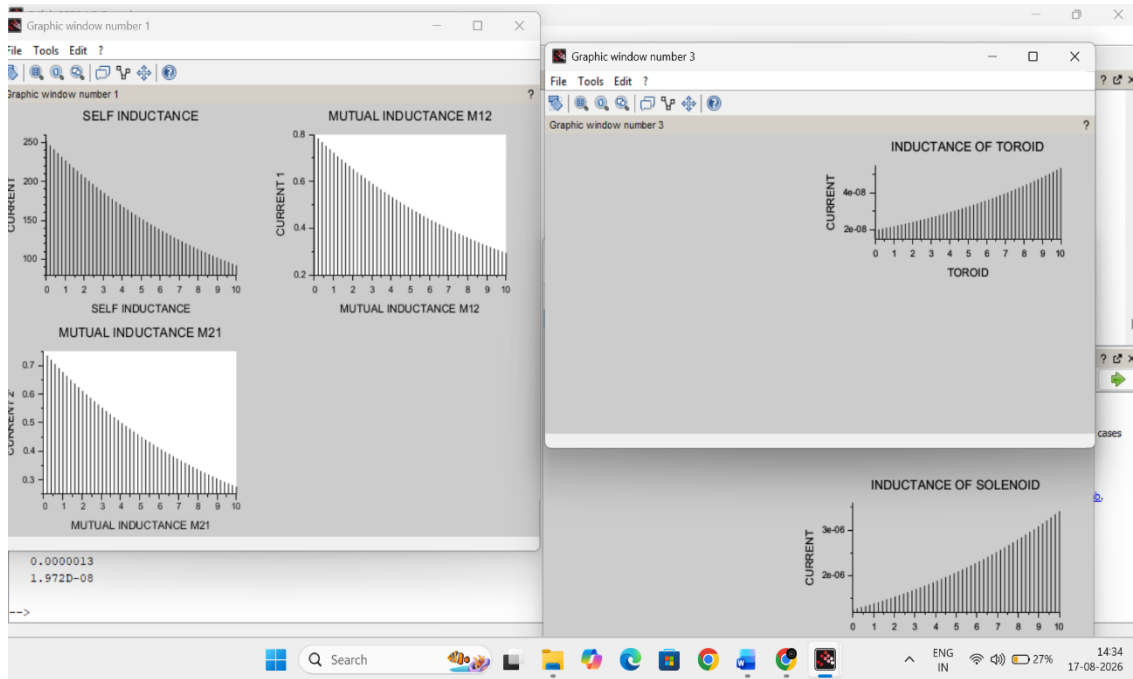
The screenshot displays the SCILAB 6.0.1 environment. The left window is the 'Console', showing the execution of a script with user inputs and calculated outputs. The right window is the 'Editor', showing the source code of the script.

Console Output:

```
Enter the phi value:3.14
Enter the current flowing in inductance:5
    "Self inductance:"
    251.2
Enter the primary turn:100
Enter the secondary turn:200
Enter the  $\phi_{12}$ : $2 \times 10^{-2}$ 
Enter the  $\phi_{21}$ : $3 \times 10^{-2}$ 
Enter the CURRENT 1:5
Enter the CURRENT 2:4
    "Mutual inductance M12:"
    0.8
    "Mutual inductance M21:"
    0.75
Enter the length value is:5
Enter the radius value is: $2 \times 10^{-3}$ 
    0.0000013
    1.972D-08
-->
```

Editor Code (Exp No 15.sc):

```
1 clear;
2 clear;
3 n=input("Enter the turns of self inductance:");
4 phi=input("Enter the phi value:");
5 i=input("Enter the current flowing in inductance:");
6 L=n*phi/i;
7 disp("Self inductance: ", L);
8 n1=input("Enter the primary turn:");
9 n2=input("Enter the secondary turn:");
10  $\phi_{12}$ =input("Enter the  $\phi_{12}$ :");
11  $\phi_{21}$ =input("Enter the  $\phi_{21}$ :");
12 i1=input("Enter the CURRENT 1:");
13 i2=input("Enter the CURRENT 2:");
14 M12=n1* $\phi_{12}$ /i1;
15 M21=n2* $\phi_{21}$ /i2;
16 disp("Mutual inductance M12: ", M12);
17 disp("Mutual inductance M21: ", M21);
18 LENGTH = input("Enter the length value is:");
19 Radius = input("Enter the radius value is:");
20 solenoid =  $4 \times 3.14 \times 10^{-7} \times i$ /LENGTH;
21 toroid =  $4 \times 3.14 \times 10^{-7} \times i$ /2*3.14*Radius;
22 disp([solenoid]);
23 disp([toroid]);
24
25 a=linspace(0,10,50);
26 b=L*exp(-0.1*a);
27 c=M12*exp(-0.1*a);
28 d=M21*exp(-0.1*a);
```



Exp No: 15

no of turns in self inductor
 $N = 400$

$$i = 5$$

$$n_1 = 100$$

$$n_2 = 200$$

length =

$$\text{length} = 5 \text{ cm}$$

$$i_1 = 5 \text{ A}$$

$$i_2 = 4 \text{ A}$$

$$\phi_{12} = 2 \times 10^{-2}$$

$$\phi_{21} = 3 \times 10^{-2}$$

$$\text{Radius} = 2 \text{ cm}$$

~~Example~~

Determine i) self inductance and MI ii) solenoid and toroid with different length and circumferential area

Self Inductance

$$L = n \phi_i$$

$$n = \frac{251.2}{3.14 \times 5} = 4$$

$$L = 251.2 \text{ H}$$

Mutual Inductance

$$M_{12} = \frac{N_2 \phi_{12}}{I_1}$$

$$M_{12} = \frac{200(2 \times 10^{-2})}{4}$$

$$= 1/5$$

$$M_{12} = 0.3 \text{ H}$$

$$M_{21} = \frac{N_1 \phi_{21}}{I_2}$$

$$M_{21} = \frac{100(3 \times 10^{-2})}{4}$$

$$= 3/4$$

$$M_{21} = 0.75 \text{ H}$$

Solenoid

$$L = \frac{\mu \pi \times 10^{-7} I}{L}$$

$$= \frac{4(3.14 \times 10^{-7})(5)}{5}$$

$$= 1.256 \times 10^{-6}$$

$$L = 1.256 \times 10^{-6}$$

$$L = 0.0000012$$

Toroid

$$L = \frac{\mu_0 I}{2\pi r}$$

$$L = \frac{(4\pi \times 10^{-7})(5)}{2\pi(2 \times 10^{-3})} = \frac{20 \times 10^{-7}}{4 \times 10^{-3}}$$

$$= 5 \times 10^{-4}$$

$$L = 5 \times 10^{-4}$$

$$L = \frac{4 \times 3.14 \times 10^{-7} \times 5}{2 \times 3.14 \times 2 \times 10^{-3}}$$

$$= \frac{6.28 \times 10^{-6}}{2 \times 3.14 \times 2 \times 10^{-3}} = 1.972 \times 10^{-3}$$

$$L = 1.972 \times 10^{-3}$$

Result:

Thus, determination and plotting of Inductance, Self-inductance, Mutual inductance of current carrying coil, solenoid and toroid with different (a) length and (b) circumferential area Are performed and verified successfully.